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JURASSIC AND CRETACEOUS PALEOMAGNETIC DATA FROM THE SOUTHERN ALPS (ITALY)

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Abstract. Jurassic and Cretaceous paleomagnetic data from 46 sites in the Venetian Alps, Trento Plateau and SE Lombardian Basin have been compiled. Nine of these site mean directions have been previously published. The sites can be divided into four age groups (Senonian, Hauterivian/Barremian, Kimmeridgian/Tithonian, and Callovian/Oxfordian). For three of the age groups, positive regional fold tests indicate (at the 99% confidence level) that the characteristic magnetization components predate the deformation. Structural mapping in the Venetian Alps allowed us to assign sites to individual thrust sheets, and we observe no significant relative rotation of thrust sheets in the nappe pile. Indeed, we observe remarkable consistency in mean direction throughout the entire sampling area for each of the four age groups. The Senonian and Hauterivian/Barremian pole positions lie close to coeval poles on African apparent polar wander paths (APWPs). This result is consistent with previous analyses of middle and Late Cretaceous data from the Venetian Alps. The new Jurassic poles deviate significantly from recently published African APWPs, reflecting either the poor definition of the African APWP in the Late Jurassic or about 20° anticlockwise rotation of the Southern Alps relative to Africa during Early Cretaceous time. The former of the two possibilities is favored by a new Late Jurassic/Early Cretaceous pole from North America, and by Early Permian paleomagnetic data from the Southern Alps which are consistent with coeval African data, implying that the movement of the Southern Alps was similar to that of Africa since Early Permian.

INTRODUCTION

The Upper Cretaceous "Scaglia" limestones exposed in the Venetian Alps (Figures 1 and 2) were a focus for paleomagnetic work in the late 1970s [Channell and Tarling, 1975; VandenBerg and Wonders, 1976, 1980; Channell et al., 1979a; Lowrie et al., 1980; Channell and Medizza, 1981]. In addition, several paleomagnetic studies are available from Permian volcanics [Zijderveld and DeJong, 1969; Zijderveld et al., 1970; Manzoni, 1970; Heiniger, 1979] and Triassic red sandstones [Manzoni, 1970] in the Dolomites and the northern part of the Lombardian Basin. Recent reviews of Southern Alpine paleomagnetic data [e.g., Lowrie, 1986; Channell, 1986; Heller et al., 1989] have concluded that the data imply little or no rotation of the Southern Alps relative to Africa at least since Early Cretaceous. The better quality Southern Alpine Permo-Triassic data are from Lower Permian porphyries [Zijderveld et al., 1970; Heiniger, 1979] and these poles plot close to Early Permian poles from Morocco and Algeria (Figure 3), implying that the Southern Alps rotated with Africa since Permian time.

The initial objective of our study was to quantify the relative rotations of thrust sheets in the Venetian Alps and to match the shape of the apparent polar wander paths (APWP) from individual thrust sheets to the African APWP. The Venetian Alps are a fold and thrust belt exposing sediments deposited in and around the Mesozoic Belluno Basin (Figures 1 and 2) and are part of the Southern Alps which are thought to have experienced dextral transpression during Tertiary deformation [Laubscher, 1983]. Large-scale clockwise rotations, such as those documented in western Sicily [Channell et al., 1990a], have been found in thrust belts characterized by dextral transpression, and if such rotations can be documented, they can be taken into account when restoring the thrust belt.

We present 34 new site mean directions from the Southern Alps. These data are augmented by 12 site mean directions which have been previously published (marked by superscripts in Table 1). Of the 34 new site mean directions, five (Val del Mis, Quero, Frisoni, Xausa, Capriolo) have yielded magnetostratigraphies published by Channell et al. [1987] and Channell and Grandesso [1987].

GEOLOGICAL SETTING

The Jurassic and Cretaceous sediments of the Southern Alps are generally considered to have been deposited on the southern passive continental margin of the Mesozoic Tethys [Bernoulli and Jenkyns, 1974; Bernoulli et al., 1979; Winterer and Bosellini, 1981]. Four major Jurassic/Cretaceous paleogeographic zones separated by approximately N-S oriented normal faults are recognized in the Southern Alps (Figure 1). From east to west these zones are the Friuli Platform, the Belluno Basin, the Trento Plateau, and the Lombardian Basin. This topography developed during Permian and Mesozoic rifting and influenced continental margin sedimentation particularly during the Jurassic and Cretaceous. In Late Jurassic and Early Cretaceous time, reef limestones characterized the Friuli Platform. Limestones composed of resedimented shallow-water debris with in situ and resedimented pelagic material were deposited in the Belluno Basin, which was essentially a slope joining the Friuli Shelf and the relatively deep Trento Plateau [Bosellini et al., 1980; Winterer and Bosellini, 1981]. The Garda Escarpment, the western boundary of the Trento Plateau, was a zone of normal faulting from Liassic through Early Cretaceous time [Castellarin, 1972; Doglioni and Bosellini, 1987]. Red nodular limestones (Ammonitico Rosso) were deposited on the Trento Plateau and in the Belluno Basin during the middle and Late Jurassic. The Ammonitico Rosso is subdivided into "Ammonitico Rosso superiore" (Kimmeridgian /Tithonian) and "Ammonitico Rosso inferiore" (Bajocian-Callovian/Oxfordian). The two units are separated by the siliceous Fonzaso Formation (Callovian-Kimmeridgian) in the Belluno Basin and on part of the Trento Plateau. In the Lombardian Basin, the Jurassic is characterized by deeper-water siliceous sediments with abundant resedimented material close

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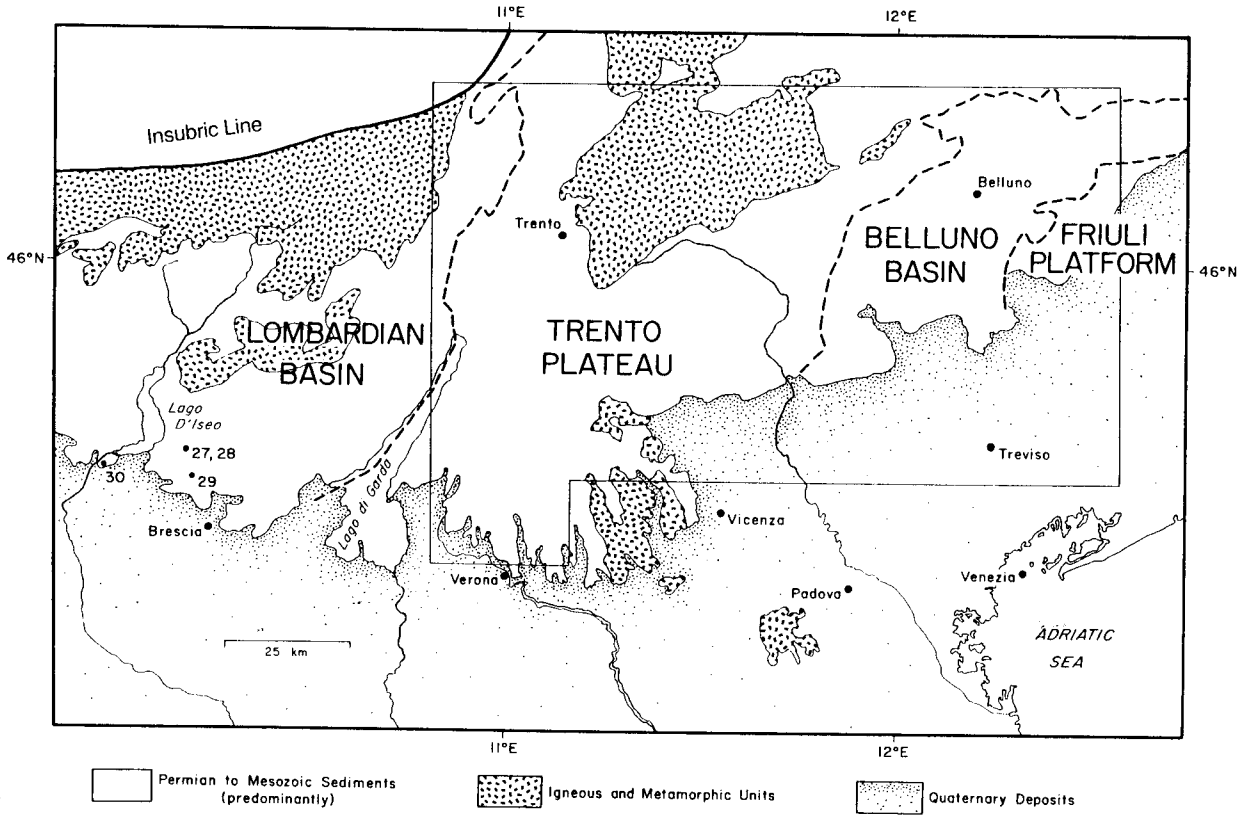


Fig. 1. Generalized geologic map of the Southern Alps indicating the Late Jurassic/Early Cretaceous paleogeographic zones. The paleomagnetic sampling sites in the Lombardian Basin (sites 27-30) are shown. Boxed area: location of Figure 2.

to fault scarps. From the late Tithonian, Southern Alpine sedimentation was more uniform as the platform/basin morphology became subdued and white/gray pelagic limestones (Maiolica Formation) blanketed much of the Lombardian and Belluno Basins as well as the Trento Plateau. Upper Cretaceous/Paleocene pale pink to red pelagic "Scaglia" limestones are also widespread, particularly in the Belluno Basin.

The Venetian Alps are a Neogene SSE vergent thrust belt. The minimum shortening in the belt is 30 km. The main overthrusts have strikes of N60°-80°E and are named, from north to south, the Valsugana line, Belluno line, Moline and Tezze line, and the Bassano/Maniago line [Doglioni, 1992] (Figures 2 and 4). Five major thrust sheets, delimited by these tectonic lines, are named in accordance with the name of the basal thrust. The thrusts are arranged in an imbricate fan geometry with a frontal triangle zone bounded by transfer faults. The principal mechanisms of shortening are fault-propagation folding and fault-bend folding [Suppe, 1983; Suppe and Medwedeff, 1990]. The Jurassic paleotectonic normal faults, which bounded the paleogeographic zones of the Mesozoic continental margin, trend N10°W-N10°E. These faults controlled facies and thickness variations in the sediment sequence which, in turn, controlled undulations in fold axes and thrust fault orientation [Doglioni, 1992].

MAGNETIC PROPERTIES

Magnetic data from four pelagic limestone units: Scaglia (Upper Cretaceous/Paleocene), Maiolica (Lower Cretaceous), and "Ammonitico Rosso superiore" (Upper Jurassic) and "Ammonitico Rosso inferiore" (Middle Jurassic), are discussed in this paper. The magnetic properties of the Scaglia and Maiolica limestones have been the subject of numerous papers [e.g., Lowrie and Alvarez, 1977; Channell and Medizza, 1981; Channell et al., 1982; Lowrie and Heller, 1982; Lowrie and Alvarez, 1984; Cirilli et al., 1984; Lowrie and Channell, 1984]. The characteristic magnetization component in the Scaglia and Maiolica limestones is carried by fine grained low-Ti magnetite. In the Scaglia, particularly in the reddish variety (Scaglia rossa), hematite contributes to the high unblocking temperature magnetization component. The magnetic properties of the Ammonitico Rosso have received less intensive study. However, available data indicate that the natural remanence (NRM) is often dominated by magnetite [Ogg, 1981; Channell et al., 1990b] with authigenic hematite, probably of early diagenetic origin, contributing in some reddened samples [Lowrie and Heller, 1982; Channell and Grandesso, 1987].

All samples analyzed in this study were treated using progressive thermal demagnetization. Representative orthogonal

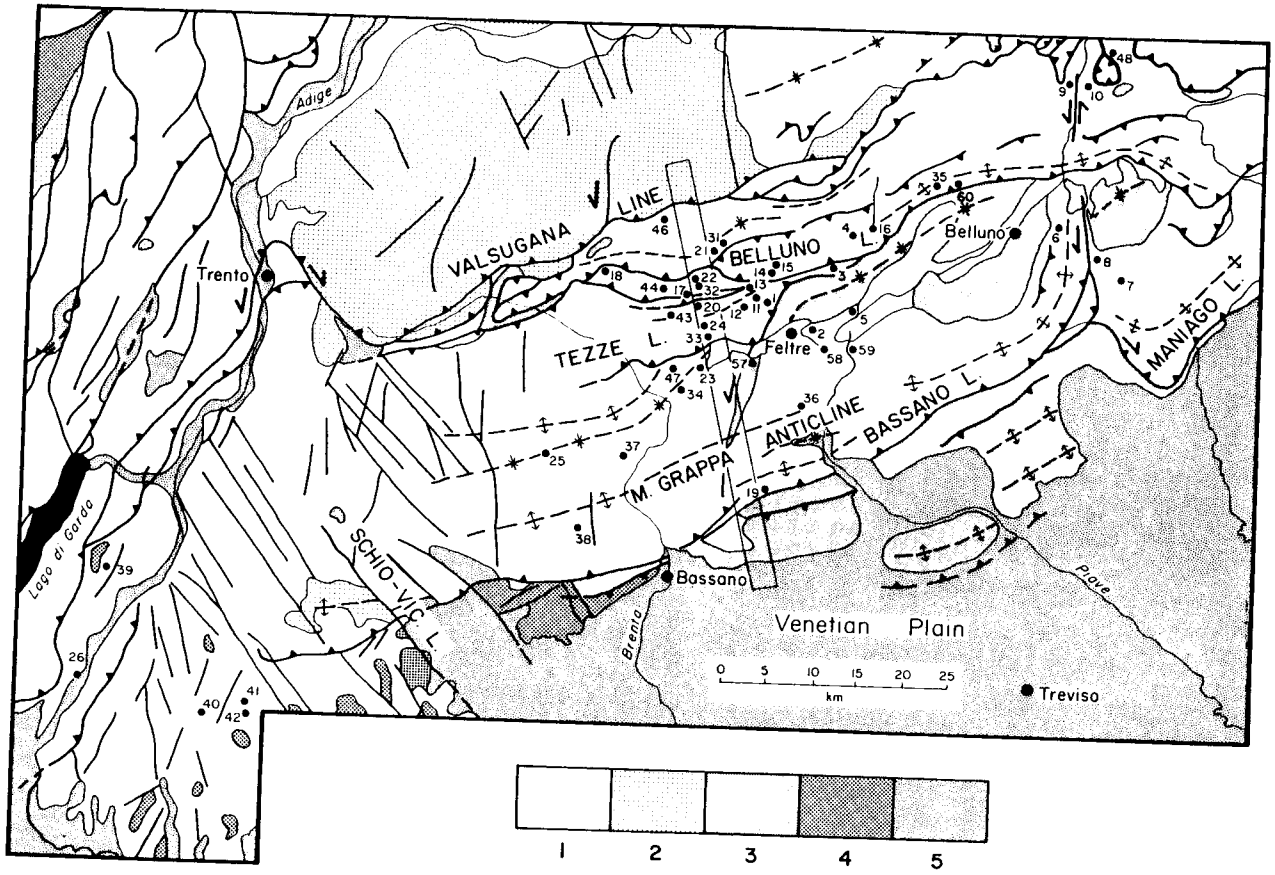


Fig. 2. Sampling site locations in the Venetian Alps (Belluno Basin and Trento Plateau paleogeographic zones). Key: 1, Upper Permian and Mesozoic sedimentary cover; 2, Hercynian crystalline basement and Permian ignimbrites; 3, Tertiary sediments (flysch and molasse); 4, Triassic and Tertiary volcanics; 5, Quaternary. Boxed area: location of cross section (Figure 4).

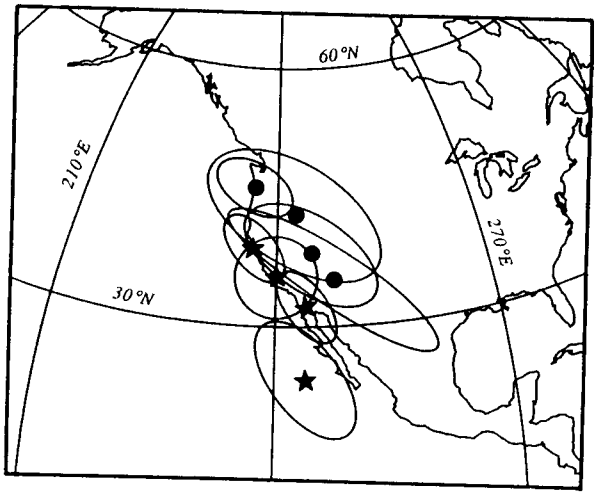


Fig. 3. Early Permian pole positions from the Southern Alps (solid circles) [Zijderveld et al., 1970; Heiniger, 1979] compared to Early Permian poles from Algeria and Morocco (stars) [Daly and Pozzi, 1976; Martin et al., 1978; Daly and Irving, 1983]. Ovals of 95% confidence are indicated.

projections of thermal demagnetization data are shown for the Scaglia (Figure 5), Maiolica (Figure 6) and Ammonitico Rosso (Figure 7). Maximum unblocking temperatures for the white/gray Maiolica limestones do not exceed 575°C (Figure 6). The maximum unblocking temperatures for the pink-colored Scaglia (Figure 5) and Ammonitico Rosso limestones (Figure 7) often exceed 600°C indicating that hematite contributes to the remanence, particularly in the more deeply reddened samples. Alternating field (AF) demagnetization of the Scaglia and Ammonitico Rosso (Figure 8) indicates that the characteristic magnetization component can be more or less resolved by alternating field demagnetization. However, the characteristic component often cannot be fully demagnetized by peak alternating fields of 100 mT, suggesting that hematite contributes to the remanence. Comparison of magnetization component directions resolved by thermal and alternating field demagnetization suggests that the characteristic component is resolved over a wide coercivity range (35-100 mT) and unblocking temperature range (300°-640°C). This observation is consistent with the conclusions of Channell et al. [1982], who maintained that the magnetization carried by hematite in Scaglia limestones was acquired during early diagenesis and has the same direction as the magnetite component except in the immediate vicinity of reversal boundaries.

TABLE 1. Paleomagnetic Data From the Belluno Basin (Venetian Alps), Trento Plateau, and SE Lombardian Basin

	Location	N	In situ		Tilt corrected		k	α_{95}
			Declination	Inclination	Declination	Inclination		
			Scaglia Rossa (Senonian)					
1	Facen Belvedere	10	346.5	-35.6	351.6	40.3	128.2	4.3
2	Feltre	10	357.0	70.8	348.3	48.2	177.4	3.6
3	Soranzen	10	350.4	-12.7	356.3	36.9	52.0	7.2
5	Busche bridge	10	356.6	35.8	354.5	18.3	236.9	3.1
6	Ponte nelle Alpi	10	210.1	-55.2	179.6	-38.2	247.5	3.1
7	Tambre d'Alpago	9	199.5	-45.6	182.1	-42.6	40.8	8.2
8	Lago S. Croce	10	158.9	-31.1	171.4	-32.8	61.2	6.2
9	Longarone	10	321.0	28.7	332.8	37.9	109.0	4.6
10	Lago del Vajont	13	332.1	37.2	345.0	38.5	91.1	4.4
11	Col Melon	11	334.7	45.2	339.3	41.7	66.9	5.6
12	Le Buse	10	355.8	43.1	352.5	34.0	298.1	2.8
13	Croce D'Aune	11	166.0	-61.5	160.5	-42.0	76.9	5.2
17	Moline	9	123.1	-68.1	163.1	-44.0	56.4	6.9
18	Castel Tesino[a]	106	344.0	52.0	354.1	43.7	12.4	4.1
19	Possagno (Carcoselle)[a]	114	345.0	14.0	343.8	42.5	13.1	3.8
20	Moline-Ponte della Serra[a]	73	335.0	68.0	344.9	38.4	11.9	5.0
43	Lamon	14	335.4	55.8	339.5	39.2	225.3	2.7
44	Lamon	12	346.6	-31.2	348.6	46.0	158.4	3.5
46	Passo Broccon	11	358.9	47.9	351.7	43.9	115.6	4.3
47	Mellame (Arsie)	14	149.3	-42.0	153.3	-29.3	69.1	4.8
57	Seren del Grappa	11	324.9	23.5	338.7	38.9	187.5	3.3
60	Val del Mis-Vedana	10	156.0	63.0	156.5	-22.0	96.0	5.0
	Quero[b]	*10*			*342.7*	*50.9*		*5.3*
	Quero[b]	*10*			*344.2*	*43.3*		*5.3*
	Schio[c]	*12s*			*343.0*	*41.4*	*33.8*	*7.5*
			Upper Maiolica (Hauterivian-Barremian)					
21	Monte Vederna	11	313.7	42.5	323.6	34.1	65.4	5.7
22	Val Rosna	9	314.6	68.0	329.7	56.7	29.8	9.6
23	Fonzaso (Frassene)	7	327.5	41.9	324.2	32.9	73.5	5.7
24	Moline-Ponte della Serra[d]	82	298.0	69.0	321.7	38.1	6.0	7.0
25	Gallio[a]	25	332.3	40.5	329.4	34.7	50.9	4.1
26	Spiazzi[a]	53	336.1	56.5	321.1	46.3	5.5	9.2
27	Polaveno[e]	367	332.8	20.2	322.0	36.1	24.6	1.5
28	San Giovanni[e]	86	333.1	21.4	323.4	36.0	32.5	2.7
29	Pie' del Dosso[e]	38	16.4	75.2	306.1	40.8	30.1	4.3
30	Capriolo	137	328.7	65.6	321.4	38.4	7.2	4.8
			Ammonitico Rosso (Kimmeridgian-Tithonian)					
4	Cesiomaggiore	9	316.1	-18.1	315.0	28.5	95.5	5.3
31	Monte Vederna	8	314.6	46.1	324.2	32.3	100.5	5.6
32	Val Rosna	12	320.4	37.1	321.8	33.2	92.2	4.5
33	Ponte della Serra	9	300.9	37.6	306.9	23.8	27.9	9.9
34	Lago d'Arsie (Rocca)	11	294.3	43.9	302.3	32.6	109.2	4.4
35	Val del Mis	121	306.8	-33.8	309.0	30.2	8.2	4.8
36	Quero	77	321.1	-1.3	311.7	35.2	7.5	6.3
37	Frisoni	79	312.7	-21.0	312.7	29.7	12.3	4.7
38	Xausa	87	286.1	-36.5	296.1	23.6	12.7	4.4
39	Pra da Stua	43	308.2	53.3	306.2	38.4	11.8	6.6
48	Val Zemola	6	297.0	8.9	302.8	26.5	34.4	11.6
	Monte Grappa[b]	*10*			*328.5*	*45.3*		*3.0*
	Trento Sites[f]	*12s*			*306.0*	*33.9*	*37.8*	*7.2*

TABLE 1. (continued)

	Location	N	In situ Declination	In situ Inclination	Tilt corrected Declination	Tilt corrected Inclination	k	α95
			Ammonitico Rosso (Callovian-Oxfordian)					
40	Piccola Mantova II[g]	107	310.3	38.2	305.9	36.5	15.7	3.6
41	Valle delle Sfingi[g]	36	308.2	43.0	308.2	43.0	9.9	8.0
42	Camposilvano[g]	20	313.0	45.6	315.8	42.8	6.2	14.3
	Mean (Scaglia Rossa)	22	343.9	35.2			5.0	15.5
					346.6	38.5	68.7	3.8
		25			(346.3)	(39.3)		
		Pole:	Latitude	63.4°N	Longitude	220.2°E	dp=2.7	dm=4.5
	Mean (upper Maiolica)	10	328.4	51.0			14.0	13.4
					322.2	39.5	87.3	5.2
		Pole:	Latitude	51.5°N	Longitude	257.0°E	dp=3.7	dm=6.2
	Mean (upper Amm. Rosso)	11	307.5	11.0			5.3	21.9
					309.7	30.6	89.1	4.9
		24			(308.5)	(32.9)		
		Pole:	Latitude	39.1°N	Longitude	263.3°E	dp=3.0	dm=5.4
	Mean (lower Amm. Rosso)	3	310.5	42.3			382.9	6.3
					309.8	40.8	226.6	8.2
		Pole:	Latitude	44.0°N	Longitude	269.8°E	dp=6.0	dm=10.0

Site location given in Figures 1 and 2. The data are divided into four age groups. N refers to the number of samples per site unless the number is followed by "s" in which case the number refers to the number of sites. Published data are indicated by an asterisk, and those for which prestructural tilt correction data are unavailable are italicized. Pole positions are computed for mean directions which exclude the italicized data. Age group mean directions including the italicized data (giving equal weight to sites) are given in parentheses. Statistical parameters after Fisher [1953].

[a]Data are from Channell [1986].

[b]Data are from VandenBerg and Wonders [1976].

[c]Data are from Channell and Tarling [1975].

[d]Data are from Lowrie et al. [1980].

[e]Data are from Channell and Erba [1992].

[f]Data are from Ogg [1981].

[g]Data are from Channell et al. [1990b].

The demagnetization ranges required to resolve the characteristic high unblocking temperature magnetization components were determined by choosing the appropriate straight line segment from orthogonal projections. The standard three-dimensional least squares fitting technique, referred to as principal component analysis [Kirschvink, 1980], was used to compute the component directions. The component directions for each site were then combined using Fisher [1953] statistics (Table 1).

ANALYSIS OF DIRECTIONAL DATA

The sites were subdivided into three groups based on age: the 22 Scaglia sites are from the upper part of the formation and can be bracketted as Senonian in age, the 10 Maiolica sites come from the upper part of the Maiolica Formation and can be bracketed as Hauterivian/Barremian. Eleven sites are from the Kimmeridgian/Tithonian "Ammonitico Rosso superiore", with three sites from the Callovian/Oxfordian "Ammonitico Rosso inferiore".

We observe good grouping of site mean directions within each age group (Table 1, Figures 9 and 10). The grouping is such that there has apparently been no significant relative rotation of thrust sheets in the Venetian Alps. Data not included in the Figures 9 and 10 are italicized in Table 1. They are not included in the figures because the original publications do not give the site mean directions prior to structural tilt correction. Included in this category are 12 sites from the Trento Plateau from Ogg [1981]. The mean directions with inclusion of these data (given in parentheses in Table 1) are not significantly dif-

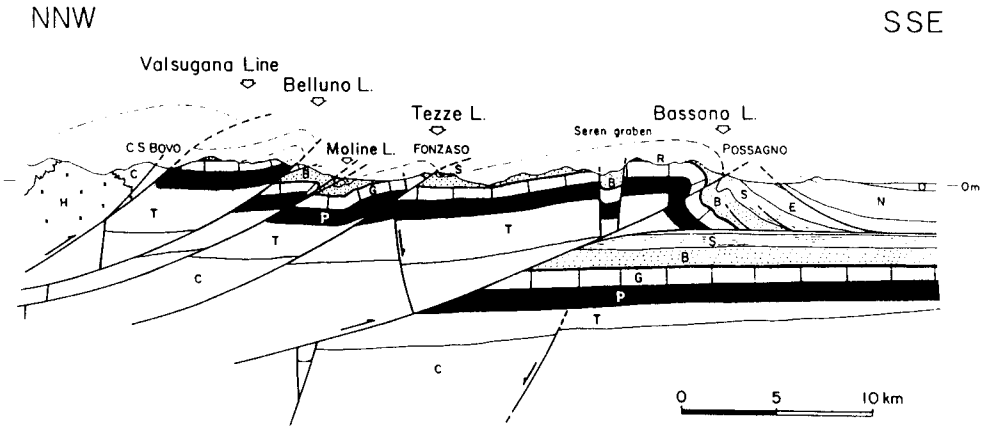


Fig. 4. Cross-section across part of the Venetian Alps. Location given by box in Figure 2. Legend: C, crystalline basement; H, Late Hercynian granite; T, Upper Permian, Lower and Middle Triassic; P, Upper Triassic (Dolomia Principale); G, Liassic platform facies (Calcari Grigi) gradually passing southward to Liassic-Dogger basinal facies in the Venetian Plane (Soverzene Fm., Igne Fm., Vajont Limstone); R, Dogger-Malm basinal facies (Ammonitico Rosso and Fonzaso Fm.); B, Lower Cretaceous (Maiolica Fm.); S, Upper Cretaceous (Scaglia rossa); E, Paleogene; N, Late Oligocene - Neogene molasse; Q, Quaternary [after Doglioni, 1992].

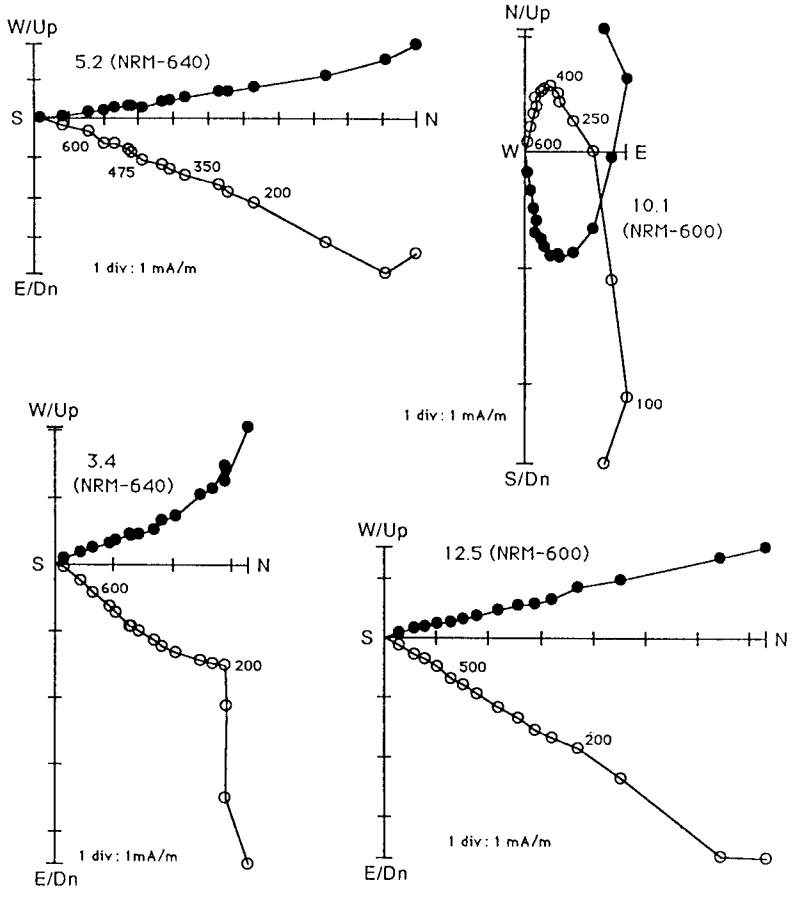


Fig. 5. Orthogonal projection of thermal demagnetization data from Senonian Scaglia limestones, with structural tilt correction. Open and closed symbols represent projection on the vertical and horizontal planes, respectively. The demagnetization range is indicated for each plot. Demagnetization temperatures are given in degrees celsius. The samples are identified by a site number (see Table 1) followed by a sample number.

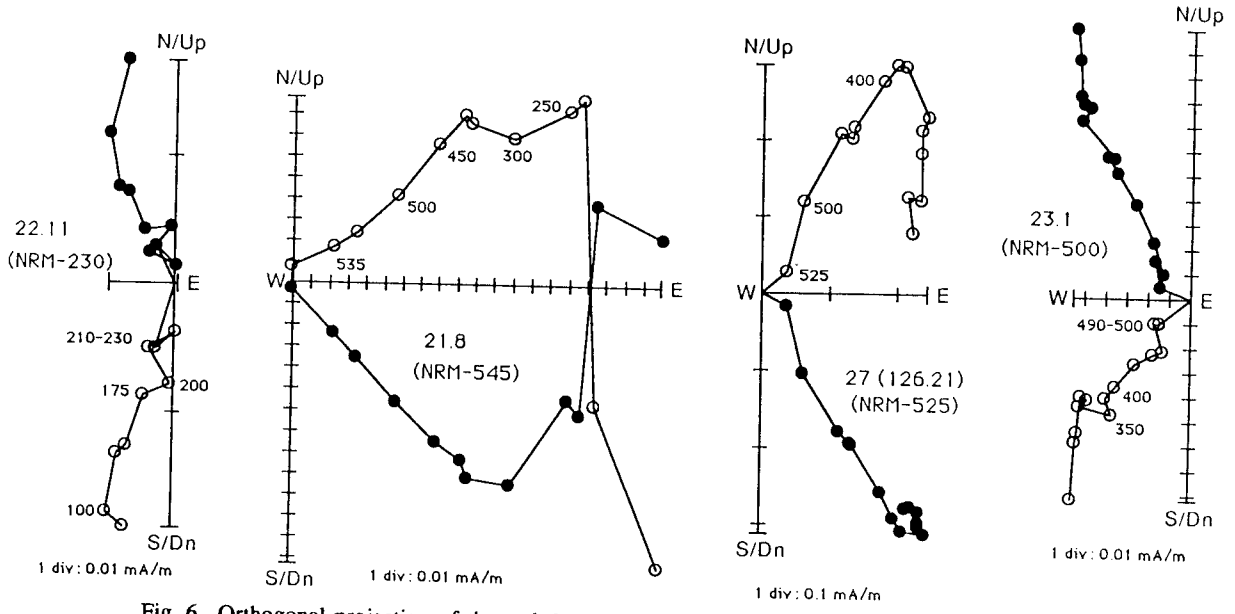


Fig. 6. Orthogonal projection of thermal demagnetization data from Hauterivian/Barremian Maiolica limestones, with structural tilt correction. Symbols are the same as in Figure 5.

ferent from the mean directions computed without their inclusion. Regional fold tests are available for three of the four age groups. All are positive at the 99% confidence level (Table 1, Figures 9 and 10) using the criteria of McElhinny [1964] or McFadden [1990]. There is insufficient variation in the tilt of the beds to conduct a fold test for the Callovian/Oxfordian sites.

There is no evidence for significant relative rotation of thrust sheets during deformation in the Venetian Alps, although coeval dextral transpression is well documented along E-W segments of the Insubric Line to the north [Laubscher, 1983; Schmid et al., 1987; Zingg and Hunziker, 1990].

APPARENT POLAR WANDER PATHS

Comparisons of Southern Alpine paleomagnetic data with Africa have been stymied by uncertainties in the African APWP. Although the situation has improved considerably in recent years, there is still a severe paucity of paleomagnetic data from Africa, particularly for the Late Jurassic/Early Cretaceous interval. In some African APWPs [e.g. Van der Voo and French, 1974; Westphal et al., 1986; Besse and Courtillot, 1991], the paucity of paleomagnetic data in this interval has been ameliorated by rotating paleomagnetic poles, using rotations corresponding to the reconstruction Pangea, to generate a

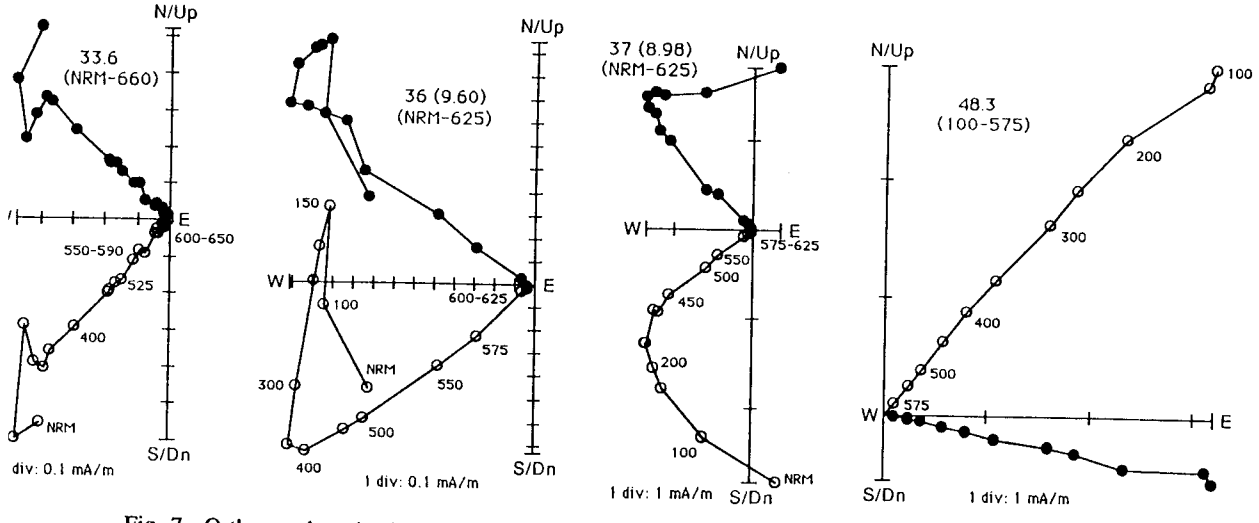


Fig. 7. Orthogonal projection of thermal demagnetization data from Kimmeridgian/Tithonian Ammonitico Rosso limestones, with structural tilt correction. Symbols are the same as in Figure 5.

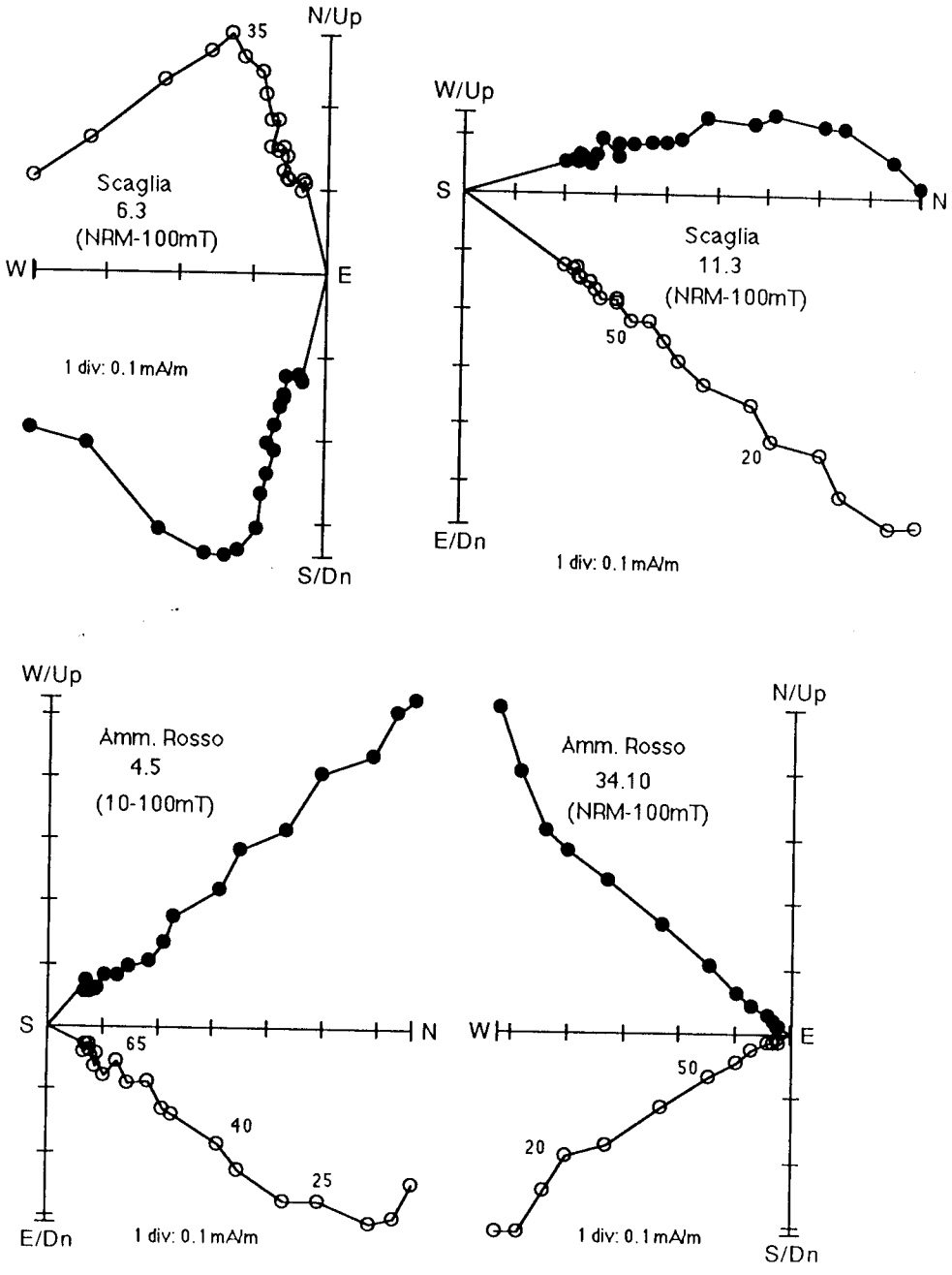


Fig. 8. Orthogonal projection of alternating field demagnetization data from Senonian Scaglia and Kimmeridgian/Tithonian Ammonitico Rosso, with structural tilt correction. The demagnetization range is given for each plot. Peak alternating fields are given in millitesla. Other symbols are the same as in Figure 5.

"master" APWP, which can then be rotated to the the coordinates of each continent. The most recent Mesozoic/Cenozoic African APWP constructed in this manner [Besse and Courtillot, 1991] is still poorly constrained in the Late Jurassic, relying heavily on relatively poor quality North American paleomagnetic data in this interval. The uncertainty is illustrated by comparing the Besse and Courtillot [1991] African APWP (Figure 11a) with selected Late Jurassic poles in African coordinates (Poles 5, 6, 7, 8, 9 in Figure 11c) [Steiner & Helsley, 1975; Van Fossen and Kent, 1990, 1992; Van der Voo, 1991]. There is considerable discrepancy in the determinations of the Late Jurassic African pole.

The Senonian (Late Cretaceous) Southern Alpine pole (LK in Figure 11b) is close to, but significantly different from, the

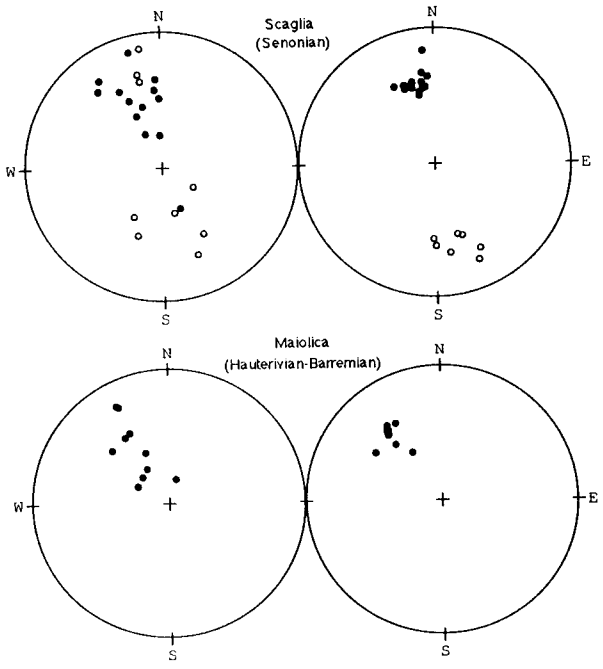


Fig. 9. Site mean directions of characteristic magnetizations (left) before and (right) after structural tilt correction, for the Senonian Scaglia and Hauterivian/Barremian Maiolica limestones. Closed and open symbols represent downward (positive) and upward (negative) inclinations, respectively. Mean directions for sites exhibiting both normal and reversed polarities are plotted with downward (positive) inclination. Fold tests are positive at the 99% confidence level for both age groups. Equal area projection.

70 Ma pole on the Besse and Courtillot path. However, the mean direction compares closely with the Late Cretaceous African pole of Hargraves [1989] (Pole 1 in Figure 11c). This may suggest that the Besse/Courtillot path is biased by Late Cretaceous data from North America and that the Hargraves pole is a better estimate for the Late Cretaceous African APWP. On the other hand, the Hauterivian/Barremian Southern Alpine pole (MK in Figure 11b) corresponds closely to the 120 Ma and 130 Ma poles on the Besse/Courtillot path. The Senonian and the Hauterivian/Barremian mean directions therefore tend to indicate that the Southern Alps have not rotated relative to Africa since Hauterivian/Barremian time. This is consistent with the conclusions of recent reviews of Southern Alpine paleomagnetic data [Lowrie, 1986; Channell, 1986; Heller et al., 1989].

The Kimmeridgian/Tithonian and Oxfordian/Callovian mean directions from the Southern Alps (Table 1) give pole positions (LJ and MJ in Figure 11b) which clearly deviate from the Besse and Courtillot [1991] African APWP. They also deviate from the Jurassic "African" poles computed by rotating Late Jurassic poles from Europe and Gondwana [Van Fossen and Kent, 1990], from North America [Steiner and Helsley, 1975], and from Lebanon and Isreal [Van der Voo, 1991] (Figure 11c). On the other hand, the Southern Alpine poles are close to the "African" pole computed from the latest Jurassic (143 Ma) New York kimberlite pole [Van Fossen and Kent, 1992] (Pole 6 in Figure 11c).

As the Southern Alpine Cretaceous mean directions appear to fit the African APWP, the discrepancy between the Southern Alpine Jurassic data and the Besse/Courtillot APWP must be due to one of the following: (1) rotation of the Southern Alps relative to Africa between Tithonian and Hauterivian/Barremian time; (2) error in the Late Jurassic part of the Besse/Courtillot APWP; (3) remagnetization in Early Cretaceous of the Southern Alpine Ammonitico Rosso. There are lines of argument which support options 1 and 2. The third option, that the Ammonitico Rosso sites were remagnetized in Early Cretaceous time, cannot be ruled out but is considered unlikely as 12 of the 14 Ammonitico Rosso sites record reversals which, for some sites, can be correlated between sections and with calpionellid biostratigraphies [see Channell and Grandesso, 1987].

Argument in favor of Early Cretaceous tectonic rotation of Adria. The apparent Early Cretaceous rotation of Adria relative to Africa is coeval with Eo-Alpine deformation in the Alps and may help to clarify a dilemma recently pointed out by Platt et al. [1989]. Several authors have documented the "African" character of Italian (mainly Cretaceous) paleomagnetic data [Channell et al., 1979b; Lowrie, 1986; Channell, 1986; Heller

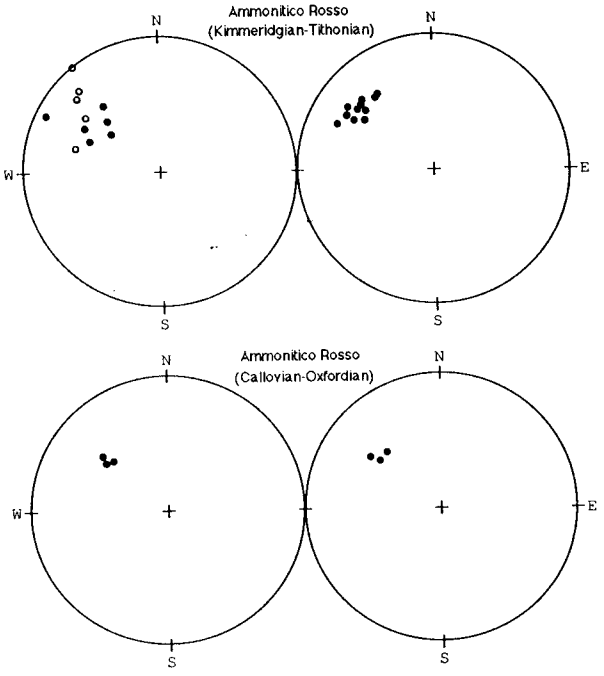
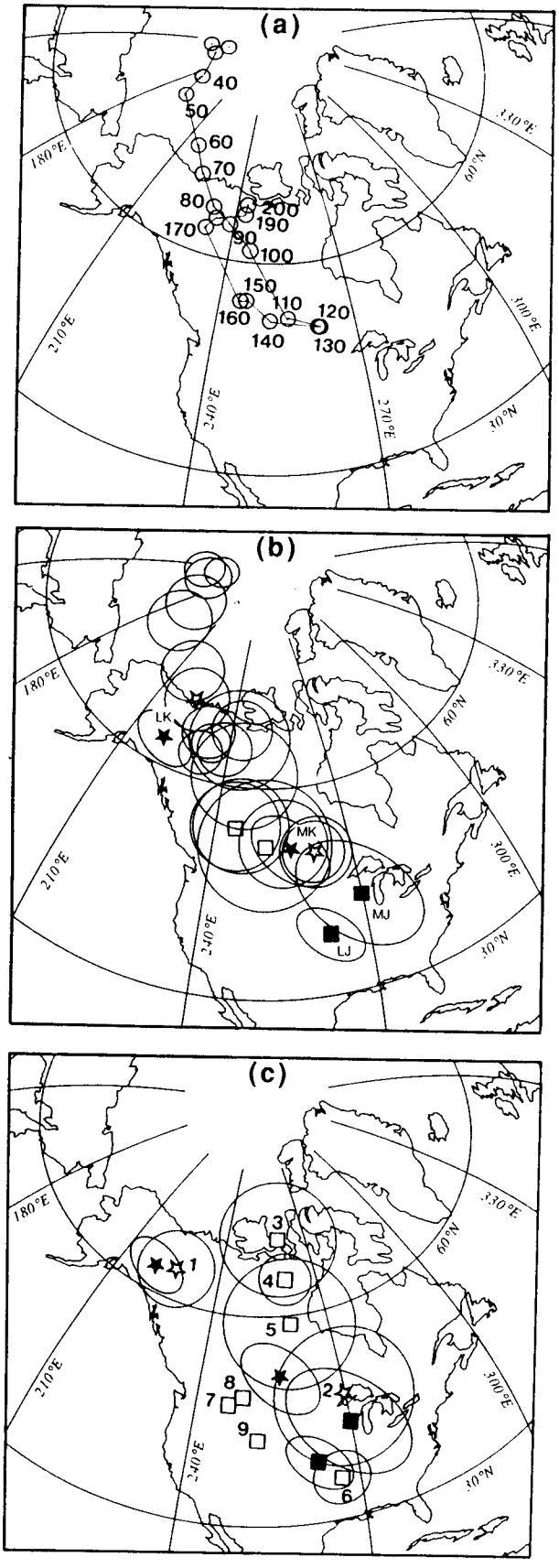


Fig. 10. Site mean directions of characteristic magnetizations (left) before and (right) after structural tilt correction (right) for the Kimmeridgian/Tithonian Ammonitico Rosso and the Callovian/Oxfordian Ammonitico Rosso. Closed and open symbols represent downward (positive) and upward (negative) inclinations, respectively. Mean directions for sites exhibiting both normal and reversed polarities are plotted with downward (positive) inclination. The fold test is positive at the 99% confidence level for the Kimmeridgian/Tithonian age group. Equal area projection.



et al., 1989], implying rotation of Adria with Africa since Early Cretaceous time. However, westward tectonic transport directions during Early Cretaceous (Eo-Alpine) deformation are not compatible with the motion history of Africa relative to Europe [Platt et al., 1989; Dewey et al., 1989], implying independent motion of Adria and Africa at this time.

Argument in favor of poor resolution of the Late Jurassic African APWP: A recently derived pole from kimberlite dykes in New York State [Van Fossen and Kent, 1992] dated at 143 Ma, close to the Jurassic/Cretaceous boundary, is supported by the presence of reversals and a baked-contact test. This pole at 58.9°N, 203.6°E is very different to age-correlative Morrison Formation poles (Upper Morrison Pole at 64.6°N, 164.2°E) [Steiner and Helsley, 1975] which anchor the North American APWP in this interval. It is interesting to note that the kimberlite pole, when rotated to African coordinates, coincides closely with the Kimmeridgian/Tithonian pole from the Southern Alps (Figure 11c). This implies that although the Early and Middle Jurassic "African" poles (poles 3 and 4 in Figure 11c) may be reliable, the Late Jurassic poles which deviate from the kimberlite pole (poles 5, 7, 8 and 9 in Figure 11c) are less dependable. The Callovian/Oxfordian pole from the Southern Alps (MJ in Figure 11b) is also a low-latitude pole lying close to the Kimmeridgian/Tithonian pole (LJ in Figure 11b), however, Callovian/Oxfordian pole is based on only three sites and has a large error associated with it.

CONCLUSIONS

Consistency of site mean paleomagnetic directions for each of the Jurassic and Cretaceous age groups indicates no significant rotation of thrust sheets in the Venetian Alps, and a uniform rotational history for the Venetian Alps, Trento Plateau and SE Lombardian Basin. Overall mean directions for Late

Fig. 11. (a) Synthetic African polar wander path (APWP) from 200 Ma to 10 Ma [Besse and Courtillot, 1991]. (b) The same APWP with 95% confidence circles. The Southern Alpine poles corresponding to the four age groups (Table 1) are indicated by closed squares for Jurassic (MJ and LJ denote middle and Late Jurassic, respectively) and closed stars for Cretaceous (MK and LK denote middle and Late Cretaceous, respectively). On the APWP, coeval middle and Late Jurassic poles and the middle and Late Cretaceous poles are highlighted by open squares (Jurassic) and open stars (Cretaceous). (c) Jurassic (closed squares) and Cretaceous (closed stars) Southern Alpine poles (as in Fig. 11b) compared with Jurassic (open squares) and Cretaceous (open stars) African poles. Key to African poles: (1) Late Cretaceous and (2) Early Cretaceous African poles from Hargraves [1989]. (3) Early, (4) Middle, and (5) Late Jurassic poles from Europe and Gondwana continents rotated to African coordinates [see Van Fossen and Kent, 1990]. (6) Latest Jurassic (143 Ma) New York kimberlite pole [Van Fossen and Kent, 1992] rotated to African coordinates. (7) Latest Jurassic Upper Morrison pole [Steiner and Helsley, 1975] rotated to African coordinates. (8) and (9) Kimmeridgian Isreali/Lebanese poles rotated to African coordinates [Van der Voo, 1991]. Confidence circles not given for poles 7-9. Rotation to African coordinates for poles 6 and 7 use Euler rotations: 66.18., -018.34, Ω= 60.2 [Klitgord and Schouten, 1986].

Cretaceous (Senonian) and Early Cretaceous (Hauterivian/Barremian) sites give pole positions which coincide closely with recently derived coeval African poles. However, Jurassic (Kimmeridgian/Tithonian and Oxfordian/Callovian) poles from the Southern Alps deviate from coeval African poles. This is probably due to uncertainties in Late Jurassic African pole positions rather than a result of rotation of the Southern Alps, relative to Africa, in Late Jurassic/Early Cretaceous time. This conclusion is consistent with the observation that available Early Permian paleomagnetic data from the Southern Alps coincide closely coeval African poles (Figure 3), suggesting a close link between the motion of the Southern Alps and Africa since Early Permian time.

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